

CSP529 Technical Writing & Publishing – W26

Practical–5

Reading Research for Evidence: Constructing an Evidence Matrix

Duration: 2 Hours **Groups:** 11 groups × 3 students

Material: Three printed research papers per group

Mode: In-class close reading and group analysis

Theme

Do not read a research paper merely to know what it says. Read it to determine what evidence it provides for the question you are investigating.

Why This Practical?

In the previous practicals, you learned to ask:

What kind of statement am I making?

What exactly is the evidence?

Does it require evidence?

Can I trust the source?

We now move from **finding and citing sources** to **reading research literature systematically**.

A research paper contains much more information than you need for any one research question.

Your task is therefore not merely to "read the paper," but to identify and extract the information that is **relevant to the question being investigated**.

Learning Outcomes

After completing this practical, you should be able to:

- read a research paper with a **specific technical question** in mind;
- identify the problem, method, evidence, findings, conditions and limitations of a study;
- distinguish a paper's general contribution from the evidence relevant to **their particular question**;
- trace extracted information to its exact location in the paper;
- compare evidence from multiple research papers systematically;
- construct an **Evidence Matrix** that can subsequently support technical synthesis and literature-review writing.

Requirement:

Each group will have:

- **one focused technical question** assigned to it [Please see the table in the accompanying document]. This question is on the area/algorithm each group has chosen in the earlier practicals.
- Based on the area and the question, each group will be finding **three research papers** relevant to that question and the area.
- **Each member of the group will be selecting one paper from the papers searched by the group.**

During the main reading exercise

No ChatGPT or other AI tools. No AI-generated summaries.

The objective is to develop your own ability to read and interrogate research literature.

Stage 1 — Begin With the Question

Before reading the papers, all three group members should discuss the assigned question.

A. Write briefly:

1. What do you understand by the focused question assigned to your group?
2. Are there any important terms in the question whose meaning needs to be clarified before reading the papers?

Do not attempt to answer the question at this stage. Read the papers to discover what evidence they provide towards answering it.

Stage 2 — Read Your Paper for Evidence

Each of you now reads one research paper.

You are **not expected to understand or summarize every part of the paper.**

Read it with one question continuously in mind:

What does this paper contribute toward answering our group's question?

While reading, annotate the printed paper.

You may use these simple markings:

Q — relevant to our Question

M — Method/approach

E — Evidence

F — Finding

C — Condition/assumption

L — Limitation

Underline or mark the relevant passage, table, equation or figure.

B. For Your Paper, Identify

1. Problem

What problem or research question does the paper address?

2. Method

How did the authors investigate it?

3. Data / Experimental Setting

What data, system, simulation, benchmark, experimental setup or other basis was used?

4. Relevant Finding

What finding is **specifically relevant to your group's focused question?**

5. Evidence

What evidence supports that finding?

Look for:

- experimental results;
- measurements;
- tables;

- graphs;
- mathematical analysis;
- simulations;
- comparisons;
- empirical observations.

6. Conditions / Assumptions

Under what conditions does the finding hold?

7. Limitations

What limitation of the work is relevant to interpreting the finding?

8. Contribution to Our Question

Complete:

"For our focused question, this paper tells us that..."

Limit this to **2–3 sentences.**

An Important Rule: Evidence Must Be Traceable

Whenever you extract an important piece of information, record **where you found it.**

Stage 3 — Each student to discuss his/her findings on the paper with other members of the group.

Do not summarize the entire paper.

Discuss only:

1. What was investigated?
2. How was it investigated?
3. What evidence is relevant to our question?
4. What does the evidence suggest?
5. Under what conditions?
6. What should we be cautious about?

The other two students may ask questions and inspect the marked portions of the paper.

Stage 4 — Construct the Evidence Matrix

This exercise is to be done together.

C. Prepare ONE common Evidence Matrix for the group.

Dimension	Paper 1	Paper 2	Paper 3
Problem/question addressed			
Method/approach			
Data/experimental setting			
Main finding relevant to our question			
Evidence supporting the finding			
Conditions/assumptions			
Limitations relevant to our question			
What this paper contributes toward answering our question			

Important

Wherever appropriate, include the **page/section/table/figure number** in the matrix. The matrix should be based on the papers—not on memory, search-engine summaries, abstracts alone, or AI-generated explanations.

Stage 5 — Looking Across the Matrix

Do not yet attempt a complete synthesis. That will be done in the next practical.

However, as a group, examine your completed matrix and answer:

D.

1. Which dimensions can reasonably be compared across all three papers?
2. Where do the papers appear to provide similar evidence?
3. Where do you notice an apparent difference?
4. Is there any important information needed for answering your question that one or more papers do not provide?

Do not try to resolve these differences yet. Simply **identify them**.

Individual Exit Exercise

Each student answers independently:

E1. Which single piece of evidence in your paper is most useful for answering the group's focused question? Why? [Give its exact location in the paper.]

E2. What important aspect of the group's question could your paper not answer?

E3. What did you understand from reading the full paper that you would probably have missed by reading only its abstract or an AI-generated summary?

Q E3 is most important. . It makes you reflect on **why you performed the manual reading exercise?**

Deliverables

At the end of the two-hour practical, each group submits:

1. **Three annotated printed research papers**
2. **One handwritten Evidence Matrix;**
3. Answers to **A and D** as a group;
4. Individual answers to **E1–E3** from all three members.

You should write in **LaTeX** all the **submissions required for this practical**.

Writing Logbook / Journal

Record:

1. What was most difficult to extract from the research paper?
2. Which section/table/figure provided the most useful evidence?
3. What did the paper's abstract fail to tell you?
4. What difference did reading with a **focused question** make?
5. What do you need to investigate further before answering the group's question?

Do not divide the final matrix into "your paper, my paper, his/her paper."

Although each of you initially reads one paper, the **Evidence Matrix belongs to the group**.

Every member should understand the important evidence extracted from all three papers.
